Data Visualizations In The Healthcare Industry

By: Vraj Patel

Dataset Used

The reason I chose to work with this dataset is because I had never worked with a healthcare related dataset before and I wanted to look at data that was foreign to me. It also had a large amount of unique variables such as “patient blood-type”, “gender”, “admission”, “discharge date”, “insurance provider”, and “billing amount”. With these variables, I thought I could capture common trends in the healthcare industry by making visuals on them.

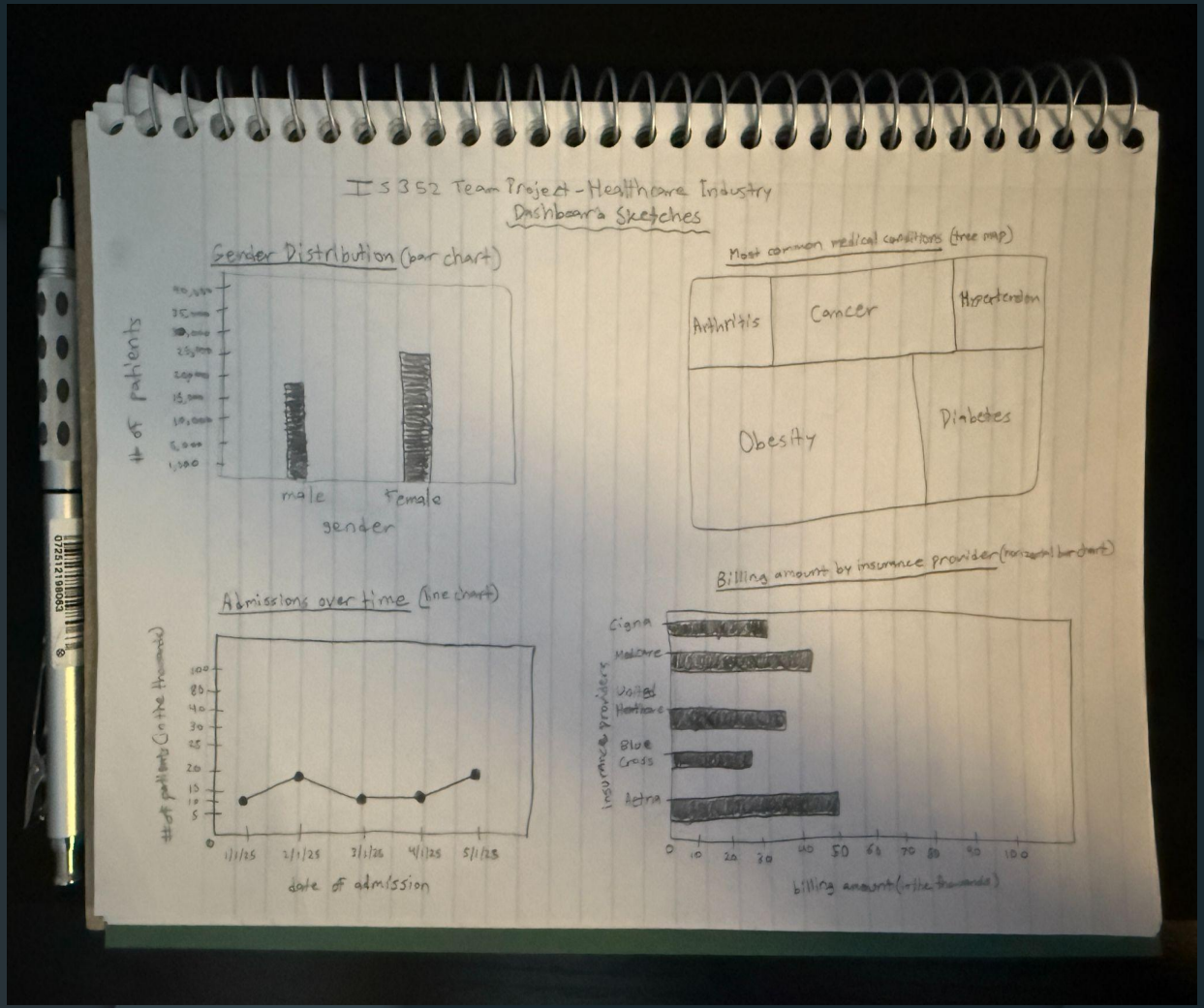
Changes made to the dataset:

- Formatted Dates to all match MM-DD-YYYY.
- Deleted Room Number column as it would not affect the visualizations I brainstormed.
- Deleted Doctor column as Doctor would not affect the visualizations as well.
- Changed all white space in variables names to underscores.
- Formatted the Billing Amount to only 2 decimal places.
- Used AI to help clean up the dataset as well. Only helped with the Name variable so that all capitalization was consistent across each entry. All other modifications were made in excel by excel formatting.

https://www.kaggle.com/datasets/prasad22/healthcare-dataset

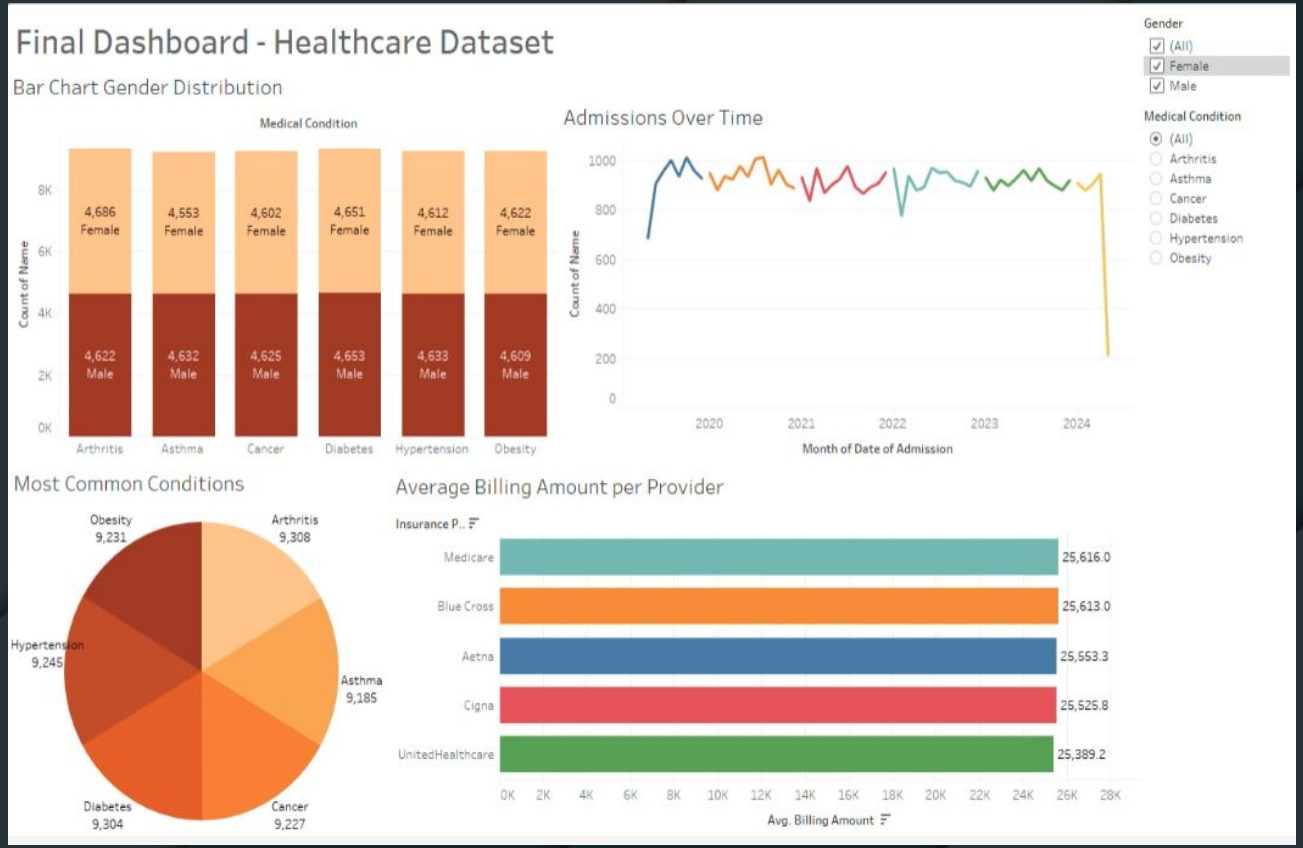
Dashboard Sketch

- I planned out this sketch by thinking about visualizations that would show trends in the data.
- I first thought about showing the gender distribution through a bar chart to show any disparity between the two.
- Next I thought that a tree map showing the most common conditions would give a detailed understanding of the disparity between the conditions.
- I also thought to use the admission date variable to show the amount of admissions over time in a MM-YYYY format to locate any spikes or dips in patients being admitted.
- Lastly, I wanted a horizontal bar chart showing the average price of a visit based on each insurance provider.



Finalized Dashboard

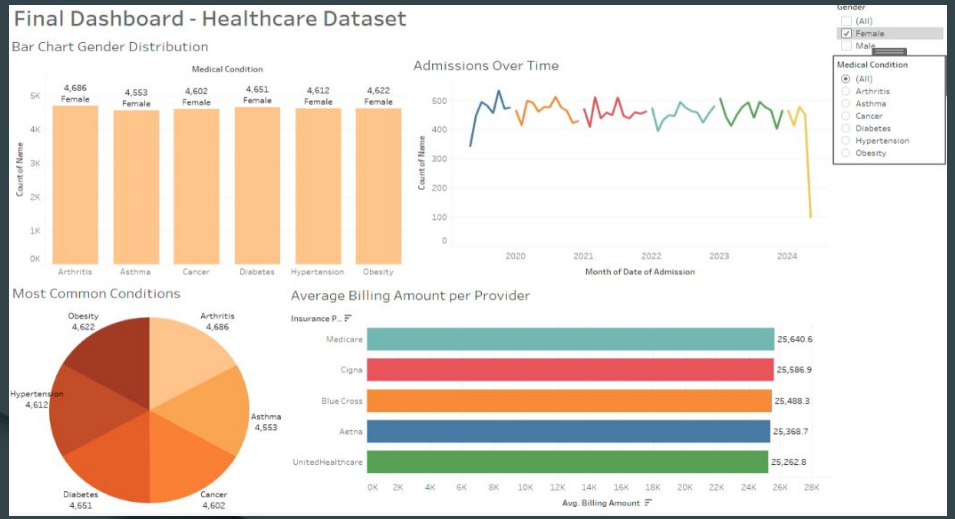
- Stacked Bar Chart showing the amount of patients with each condition separated by gender (with amount and gender title shown).
- A Line Chart showing the admissions over time in MM-YYYY format.
- Pie chart showing the most common conditions and the different colors represent the different conditions.
- The Horizontal Bar Chart on the left shows the average billing amount per healthcare provider.
- Some visualizations were altered from my sketch because I thought it gave a better representation of the data.

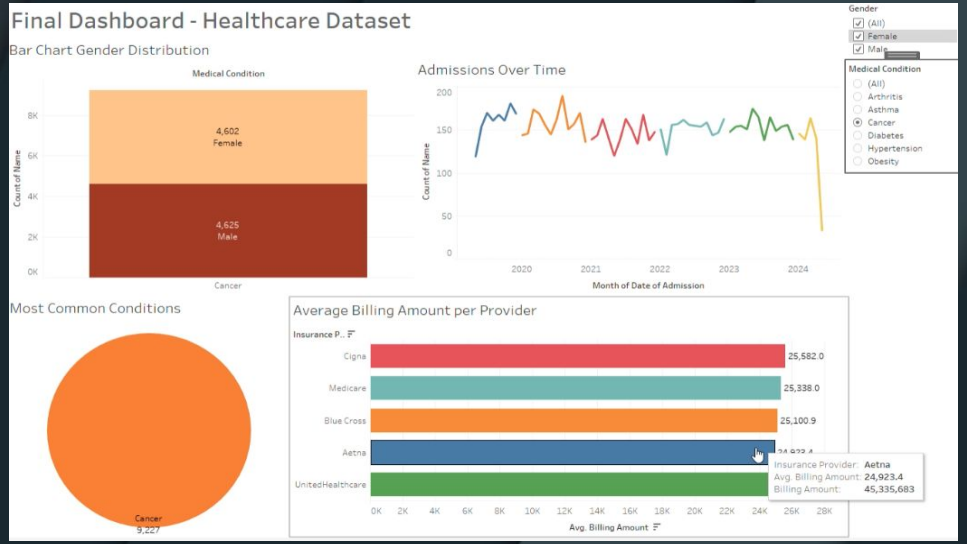


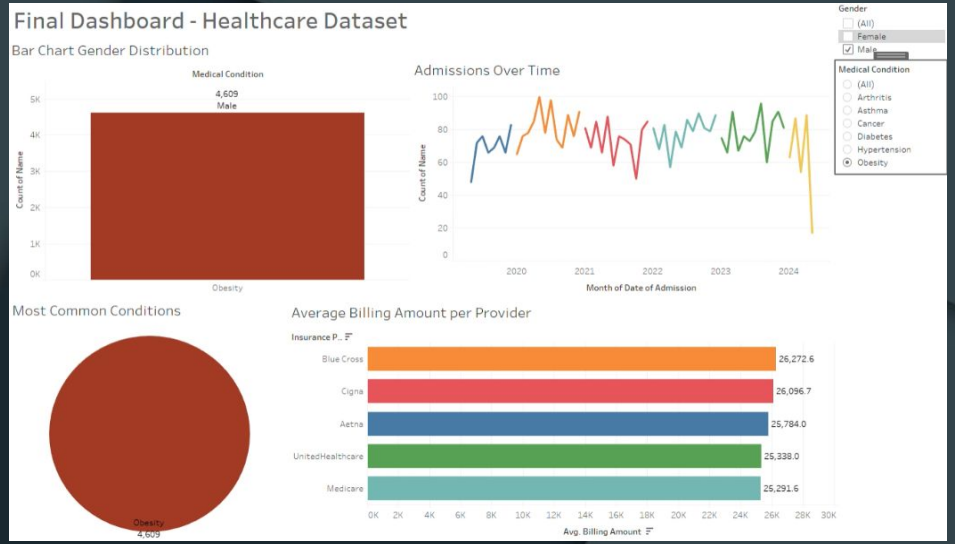
Tableau

Filters in Action

- Images showing the dashboard before and after applying filters.
- The top right dashboard shows all charts with only the female gender selected.
- The bottom left dashboard shows all genders selected but only the condition of cancer.
- The bottom right dashboard shows specifically charts based on the male gender with the condition of cancer.







Insights and Recommendations

- One of the key trends that I found was in the line chart that tracked admissions over time. With all conditions and genders selected, I noticed that there were spikes in the summer months for the amount of patients that were being admitted.
- Another Key insight that I found was that the average billing cost per each insurance provider was around the same. The difference from the cheapest to most expensive was only around $250 disparity. Medicare being the most expensive and UnitedHealthcare being the cheapest for ALL conditions. These prices change however when certain conditions are selected, but the margin still never goes over $1000.
- One other key insight I found was from the stacked bar chart. I noticed that for the conditions arthritis and diabetes, there were more female patients than male patients, and for the other 4 conditions, there were more males admitted with those conditions.

<u>Actionable steps</u>

- Look into why there is an increase of patients admitted in warmer months.
- Look into why females are getting arthritis and diabetes more than males.
- Look into how each insurance provider is charging similar amounts for specific conditions.